

Topological Convolution

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May 15, 2026

Simplex

Definition. Given a finite set of vertices $\mathcal{V}$, a **k-simplex** $\mathcal{S}^k$ is an unordered subset containing $k + 1$ elements $\{v_0, \dots, v_{K+1}\}$.

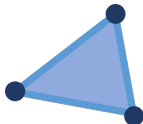
- Embedded in the Euclidean space + relationship to express closeness



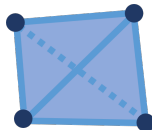
0-simplex



1-simplex



2-simplex



3-simplex

Algebraic Representation

- **Hodge Laplacians:** describe the **structure** of the whole K -simplicial complex

$$\mathbf{L}_0 = \mathbf{B}_1 \mathbf{B}_1^\top \longleftarrow \text{graph Laplacian}$$

$$\mathbf{L}_1 = \mathbf{B}_1^\top \mathbf{B}_1 + \mathbf{B}_2 \mathbf{B}_2^\top$$

$$\mathbf{L}_k = \mathbf{B}_k^\top \mathbf{B}_k + \mathbf{B}_{k+1} \mathbf{B}_{k+1}^\top \quad k = 2, \dots, K-1$$

$$\mathbf{L}_K = \mathbf{B}_K^\top \mathbf{B}_K$$

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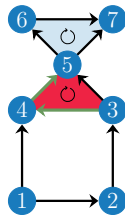
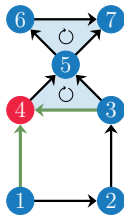
$$\mathbf{L}_k = \mathbf{B}_k^\top \mathbf{B}_k + \mathbf{B}_{k+1} \mathbf{B}_{k+1}^\top \quad k = 2, \dots, K-1$$

$$\mathbf{L}_K = \mathbf{B}_K^\top \mathbf{B}_K$$

- **Intermediate Laplacians** encode both **lower** and **upper** neighborhood **relations**

⇒ **Lower** Laplacian: $\mathbf{L}_{k,\ell} = \mathbf{B}_k^\top \mathbf{B}_k$ for lower **adjacencies**: Edges (3, 4) and (1, 4) are lower adjacent

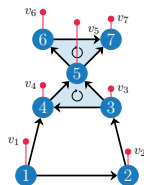
⇒ **Up** Laplacian: $\mathbf{L}_{k,u} = \mathbf{B}_{k+1} \mathbf{B}_{k+1}^\top$ for upper **adjacencies**: Edge (3, 4) and (4, 5) are upper adjacent



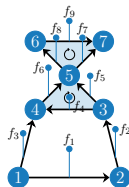
Simplicial Signals

Definition. A **simplicial signal** is a **mapping** from the simplex set the real domain $f : \mathcal{S}^k \rightarrow \mathbb{R}$.

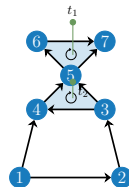
- **Simplicial signals** $\Rightarrow$ Associate a value to each simplex
 - $\Rightarrow$ collect in vector $\mathbf{x}^k = [x_1^k, \dots, x_{N_k}^k]^\top \in \mathbb{R}^{N_k}$
- **Examples:**
 - $\Rightarrow$ **Node signals:** $\mathbf{x}^0 := \mathbf{v} \in \mathbb{R}^{N_0}$ measurements of temperature, water pressure
 - $\Rightarrow$ **Edge signals:** $\mathbf{x}^1 := \mathbf{f} \in \mathbb{R}^{N_1}$ Measurement of water flow, traffic flow
 - $\Rightarrow$ **Triangle signals:** $\mathbf{x}^2 := \mathbf{t} \in \mathbb{R}^{N_2}$ EEG measurements of three voxel interaction



Node signals



Edge signals



Triangle signals

Outline: What is This Lecture About?

► How to perform convolutions with topological signals?

⇒ Topological convolution

⇒ Equivariances

⇒ Topological dictionaries

⇒ Convolution across levels

Topological Convolutions

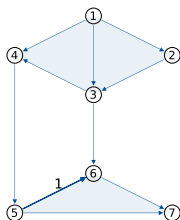
Simplicial Shift via Laplacian

- The Hodge Laplacian acts as the signal shift in the simplicial domain

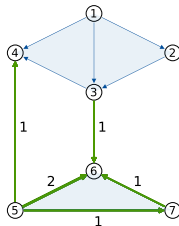
$$\mathbf{L}_1 = \mathbf{L}_{1,\ell} + \mathbf{L}_{1,u}$$

- For an edge flow $\mathbf{f}$, operation $\mathbf{L}_1$ shifts the signal $\mathbf{f}^{(1)} = \mathbf{L}_{1,\ell}\mathbf{f} + \mathbf{L}_{1,u}\mathbf{f}$

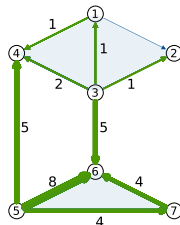
$$\Rightarrow \mathbf{L}_{1,\ell}\mathbf{f} - \text{lower neighbor shift} \quad \left[\mathbf{f}_\ell^{(1)} \right]_i = \sum_{j \in \{ \mathcal{N}_{\ell,i}^1 \cup i \}} [\mathbf{L}_{1,\ell}]_{ij} [\mathbf{f}]_j$$



$\mathbf{f}$



$\mathbf{L}_{1,\ell}\mathbf{f}$



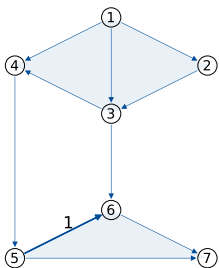
$\mathbf{L}_{1,\ell}^2\mathbf{f}$

- Two hops lower shift $\Rightarrow \mathbf{f}_\ell^{(2)} = \mathbf{L}_{1,\ell}^2\mathbf{f} = \mathbf{L}_{1,\ell}\mathbf{f}_\ell^{(1)}$

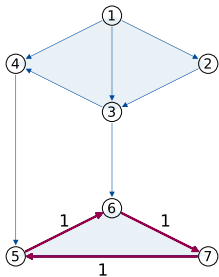
Simplicial Shift via Laplacian

► $\mathbf{L}_{1,u}\mathbf{f}$ – upper neighbor shift $\left[\mathbf{f}_u^{(1)}\right]_i = \sum_{j \in \{\mathcal{N}_{u,i}^1 \cup i\}} [\mathbf{L}_{1,u}]_{ij} [\mathbf{f}]_j$

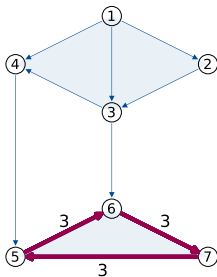
$\Rightarrow$ Two hops upper shift $\Rightarrow \mathbf{f}_u^{(2)} = \mathbf{L}_{1,u}^2 \mathbf{f} = \mathbf{L}_{1,u} \mathbf{f}_u^{(1)}$



$\mathbf{f}$



$\mathbf{L}_{1,u}\mathbf{f}$



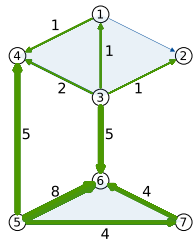
$\mathbf{L}_{1,u}^2 \mathbf{f}$

Simplicial Shift via Laplacian

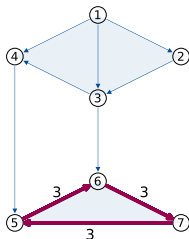
- Multi-hop simplicial signal shifts

$$\mathbf{f}^{(k)} = \mathbf{L}_{1,\ell}^k \mathbf{f} + \mathbf{L}_{1,u}^k \mathbf{f}$$

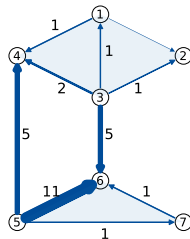
$$\Rightarrow \mathbf{L}_{1,\ell}^k \mathbf{L}_{1,u}^k = \mathbf{0} \text{ since } \mathbf{B}_k \mathbf{B}_{k+1} = \mathbf{0}$$



$$\mathbf{L}_{1,\ell}^2 \mathbf{f}$$



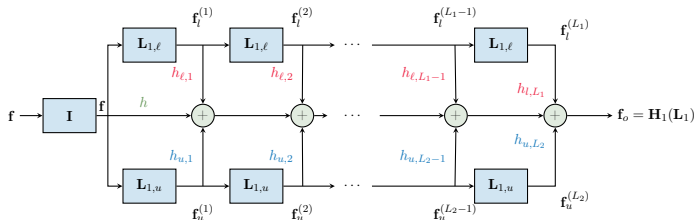
$$\mathbf{L}_{1,u}^2 \mathbf{f}$$



$$\mathbf{L}_{1,\ell}^2 \mathbf{f} + \mathbf{L}_{1,u}^2 \mathbf{f}$$

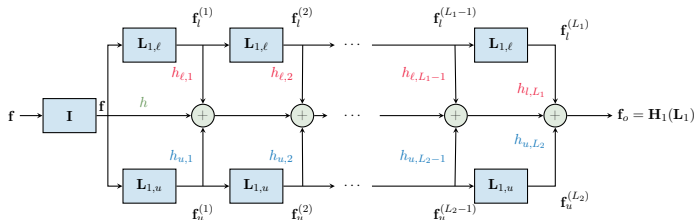
Simplicial Convolutional Filters

- Simplicial **convolutional** filtering via **shifting and sum**



Simplicial Convolutional Filters

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Definition. Simplicial convolutional filters

$$\mathbf{H}(\mathbf{L}_1) = h\mathbf{I} + \sum_{l_1=1}^{L_1} h_{l,l_1} (\mathbf{L}_{1,\ell})^{l_1} + \sum_{l_2=1}^{L_2} h_{u,l_2} (\mathbf{L}_{1,u})^{l_2}$$

- Filter output: $\mathbf{f}_o = \mathbf{H}(\mathbf{L}_1)\mathbf{f}$

Learning with Simplicial Convolutions

- The simplicial convolutional filter acts as my mapping: $\Phi(\mathbf{f}, \mathbf{L}_1, \mathcal{H}) \rightarrow \mathbf{f}_o$

$$\Phi(\mathbf{f}, \mathbf{L}_1, \mathcal{H}) = \left(h\mathbf{I} + \sum_{l_1=1}^{L_1} h_{l,l_1} (\mathbf{L}_{1,\ell})^{l_1} + \sum_{l_2=1}^{L_2} h_{u,l_2} (\mathbf{L}_{1,u})^{l_2} \right) \mathbf{f}$$

- Optimal regressor restricted to filter class

$$\mathcal{H}^* = \arg \min_{\mathcal{H}} \sum_{\mathbf{f} \in \mathcal{T}} \mathcal{L}(\Phi(\mathbf{f}, \mathbf{L}_1, \mathcal{H}))$$

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- The simplicial convolutional filter acts as my mapping: $\Phi(\mathbf{f}, \mathbf{L}_1, \mathcal{H}) \rightarrow \mathbf{f}_o$

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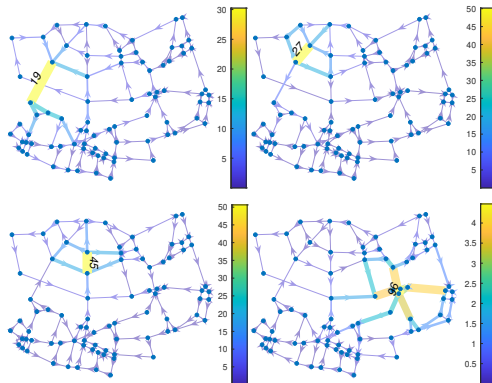
- simplicial-induced learning – $\mathbf{L}_1$
 - ⇒ parameters $\mathcal{H}$ independent on the problem dimension
 - ⇒ transferable – inductive learning
 - ⇒ $\Phi(\cdot)$ is local
 - ⇒ Low computational complexity – sparsity of the simplex

Application to *Fast* Edge PageRank

- **Problem.** Given a network **assess** the topological **importance** of an edge
⇒ **Input:** indicator edge flow **f** ⇒ **output** probability vector of edge importance **π**

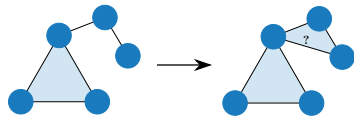
$$(\gamma \mathbf{I} + \mathbf{L}_{1,n})\boldsymbol{\pi} = \mathbf{f} \longrightarrow \boldsymbol{\pi} = (\gamma \mathbf{I} + \mathbf{L}_{1,n})^{-1}\mathbf{f}$$

- Approximate the inverse by a simplicial convolutional filter **H(L_{1,n})**



Numerical Comparison

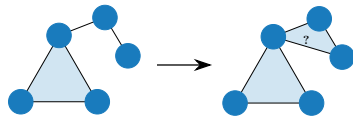
- Closure prediction
⇒ Predict the creation of social groups



Numerical Comparison

- Closure prediction

⇒ Predict the creation of social groups



Methods	2-Simplex	3-Simplex
Harm. Mean (Benson et al., 2018)	62.8 $\pm$ 2.7	63.6 $\pm$ 1.6
MLP	68.5 $\pm$ 1.6	69.0 $\pm$ 2.2
GF (Sandryhaila & Moura, 2013)	78.7 $\pm$ 1.2	83.9 $\pm$ 2.3
SCF (Yang et al., 2022b)	92.6 $\pm$ 1.8	94.9 $\pm$ 1.0
CF-SC (Isufi & Yang, 2022)	96.9 $\pm$ 0.8	97.9 $\pm$ 0.7
GNN (Gama et al., 2020a)	93.9 $\pm$ 1.0	96.6 $\pm$ 0.5

- Topological structure is relevant in practical scenarios

Learning Activity: Choosing the Right Shift Operator

In practical applications, the choice between lower-adjacency and upper-adjacency shifts determines which physical interactions are captured by a simplicial convolution.

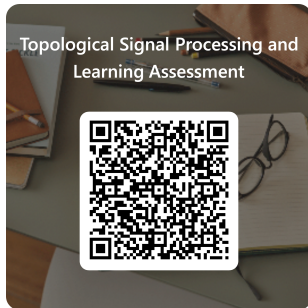
- ▶ **Q5.1:** You are modeling traffic flow on a road network where the signal is defined on road segments (edges). Two road segments should influence each other strongly if they meet at an intersection, even if they are not part of any triangle-like structure. Which design choice is most appropriate?

- ▶ **Instructions**

- ⇒ Scan the QR code to open the activity
- ⇒ Work individually first
- ⇒ Then discuss your answer in pairs or groups of 3
- ⇒ Submit your final answer in Microsoft Forms

- ▶ **Logistics**

- ⇒ **Time:** 10 min
- ⇒ **Goal:** Connect simplicial shifts to real-world flow modeling
- ⇒ **Product:** One submitted response per student + justification to your peers



<https://forms.office.com/>

Equivariances

Role of Equivariances in Topologies

What is Equivariance?

- ▶ A change in the simplicial complex's representation $\Rightarrow$ An equivalent change in the processed signal.
- ▶ Filter output depends on the intrinsic topology and physical flow, not our arbitrary conventions.

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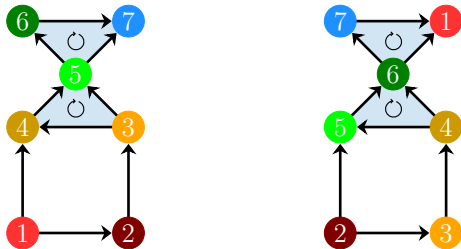
Why is it Crucial for Simplicial Complexes?

- ▶ **Handles Arbitrary Orientations:** Edge reference directions (e.g., $i \rightarrow j$ vs $j \rightarrow i$) are often assigned arbitrarily. Orientation equivariance ensures the true physical flow is preserved.
- ▶ **Ignores Simplex Indexing:** Permutation equivariance guarantees the processing is strictly independent of how we number the nodes, edges, or triangles.
- ▶ **Powerful Inductive Bias:** By embedding these topological symmetries mathematically, models (like simplicial filters) become highly data-efficient without needing to "learn" basic physical rules.

Permutation Equivariance

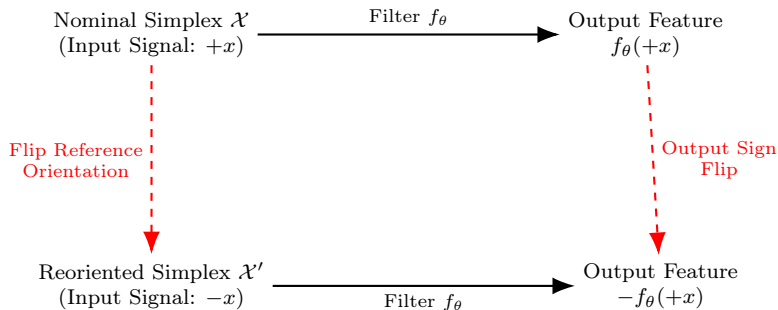
Proposition (Permutation equivariance). Consider a Hodge Laplacian $\mathbf{L}_1$ and its permuted version $\bar{\mathbf{L}}_1 = \mathbf{P}_1 \mathbf{L}_1 \mathbf{P}_1^\top$. For the simplicial signals $\mathbf{f}$ and $\bar{\mathbf{f}} = \mathbf{P}_1 \mathbf{f}$, with $\mathbf{H}(\bar{\mathbf{L}}_1)$ the permuted filter the simplicial filter outputs $\mathbf{f}_o := \mathbf{H}(\mathbf{L}_1) \mathbf{f}$ and $\bar{\mathbf{f}}_o := \mathbf{H}(\bar{\mathbf{L}}_1) \bar{\mathbf{f}}$ satisfy

$$\bar{\mathbf{f}}_o = \mathbf{H}(\bar{\mathbf{L}}_1) \bar{\mathbf{f}} = \mathbf{H}(\bar{\mathbf{L}}_1) (\mathbf{P}_1 \mathbf{f}) = \mathbf{P}_1 \mathbf{H}(\mathbf{L}_1) \mathbf{f} = \mathbf{P}_1 \mathbf{f}_o$$



- Filter output on permuted simplex $\bar{\mathbf{f}}_o$ is a permuted version $\mathbf{P}_1$ of filter output on nominal simplex $\mathbf{f}_o$

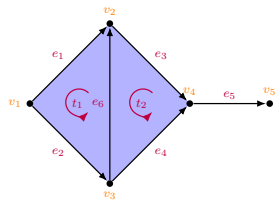
Orientation Equivariance



- Changing the reference orientation of a simplex is fundamentally a **change of basis**
- An orientation equivariant function **only changes the sign** of the output

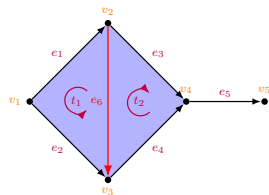
Symmetries: Orientation Equivariance

- Changing simplex orientation: it flips its relative orientation with respect to its adjacent neighbours.



$$\mathbf{B}_1 = \begin{bmatrix} -1 & -1 & 0 & 0 & 0 & \mathbf{0} \\ 1 & 0 & -1 & 0 & 0 & \mathbf{1} \\ 0 & 1 & 0 & -1 & 0 & \mathbf{-1} \\ 0 & 0 & 1 & 1 & -1 & \mathbf{0} \\ 0 & 0 & 0 & 0 & 1 & \mathbf{0} \end{bmatrix}$$

$$\mathbf{B}_2 = \begin{bmatrix} -1 & 0 \\ 1 & 0 \\ 0 & -1 \\ 0 & 1 \\ 0 & 0 \\ \mathbf{1} & \mathbf{-1} \end{bmatrix}$$



$$\bar{\mathbf{B}}_1 = \begin{bmatrix} -1 & -1 & 0 & 0 & 0 & \mathbf{0} \\ 1 & 0 & -1 & 0 & 0 & \mathbf{-1} \\ 0 & 1 & 0 & -1 & 0 & \mathbf{1} \\ 0 & 0 & 1 & 1 & -1 & \mathbf{0} \\ 0 & 0 & 0 & 0 & 1 & \mathbf{0} \end{bmatrix}$$

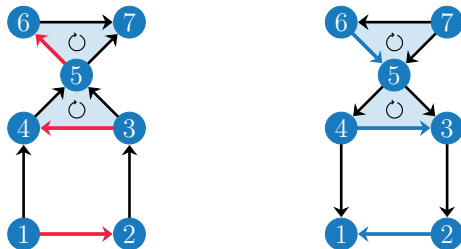
$$\bar{\mathbf{B}}_2 = \begin{bmatrix} -1 & 0 \\ 1 & 0 \\ 0 & -1 \\ 0 & 1 \\ 0 & 0 \\ \mathbf{-1} & \mathbf{1} \end{bmatrix}$$

- This amounts to flipping the sign in the corresponding rows and columns of the boundary matrices.

Orientation Equivariance

Proposition (Orientation equivariance). Consider the Hodge Laplacians $\mathbf{L}_1$ and $\bar{\mathbf{L}}_1 = \mathbf{D}_1 \mathbf{L}_1 \mathbf{D}_1$ a sequence of orientation changes. For the simplicial signals $\mathbf{f}$ and $\bar{\mathbf{f}} = \mathbf{D}_1 \mathbf{f}$, with $\mathbf{H}(\bar{\mathbf{L}}_1)$ the reorientated filter. The simplicial filter outputs $\mathbf{f}_o := \mathbf{H}(\mathbf{L}_1) \mathbf{f}$ and $\bar{\mathbf{f}}_o := \mathbf{H}(\bar{\mathbf{L}}_1) \bar{\mathbf{f}}$ satisfy

$$\bar{\mathbf{f}}_{\mathbf{o}} := \mathbf{H}(\bar{\mathbf{L}}_1) \bar{\mathbf{f}} = \mathbf{H}(\bar{\mathbf{L}}_1) (\mathbf{D}_1 \mathbf{f}) = \mathbf{D}_1 \mathbf{H}(\mathbf{L}_1) \mathbf{f} = \mathbf{D}_1 \mathbf{f}_{\mathbf{o}}$$



- Filter output on reoriented simplex $\mathbf{f}_o$ is a **reoriented** version $\mathbf{D}_1$ of filter output on nominal simplex $\mathbf{f}_o$

Learning Activity: Why Equivariance Matters

Simplicial convolutional filters should depend on the **intrinsic topology** of the complex, not on arbitrary choices such as simplex ordering or reference orientations. This motivates the need for **equivariance**.

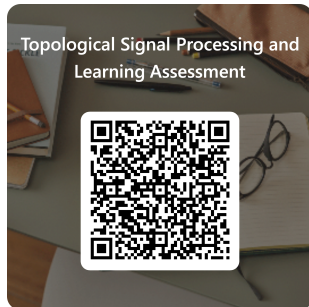
► **Q5.2:** In simplicial convolutional filtering, why is equivariance considered a crucial property?

► **Instructions**

- ⇒ Scan the QR code to open the activity
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- ⇒ Submit your final answer in Microsoft Forms

► **Logistics**

- ⇒ **Time:** 8 min
- ⇒ **Goal:** Understand why permutation and orientation symmetries are essential in topological learning
- ⇒ **Product:** One submitted response per student + justification to your peers



<https://forms.office.com/>

Dictionary Representation

Sparse Dictionary Representation

- Construct a **dictionary** that expresses the observed **signals** using only a **small number of atoms**.

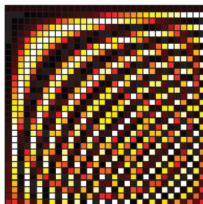
Topological
Signal

$$\mathbf{y} \in \mathbb{C}^M$$



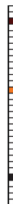
Dictionary
(change-of-basis)

$$D \in \mathbb{C}^{M \times N}$$



Sparse
Representation

$$\mathbf{x} \in \mathbb{C}^N$$



=

- **Q:** How to design and learn dictionaries for topological signals?

Topological Parametric Dictionaries

- A filter acting on edge signals is a polynomial of the Laplacians:

$$\mathbf{H}(\mathbf{L}_1) = \textcolor{green}{h}\mathbf{I} + \sum_{l_1=1}^{L_1} \textcolor{red}{h}_{\textcolor{red}{l}, \textcolor{red}{l}_1} (\mathbf{L}_{1,\ell})^{l_1} + \sum_{l_2=1}^{L_2} \textcolor{blue}{h}_{\textcolor{blue}{u}, \textcolor{blue}{l}_2} (\mathbf{L}_{1,u})^{l_2}$$

Topological Parametric Dictionaries

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- We to **know** $\mathbf{L}_{1, \ell}$, (i.e., the skeleton of the cell complex). Letting

$$\mathbf{L}_{1, u} = \sum_{j=1}^P \gamma_j \mathbf{b}_j \mathbf{b}_j^\top$$

$\Rightarrow \mathbf{b}_j \in \mathbb{R}^{N_1}, j = 1, \dots, P$, is an incidence vector associated with the j -th cycle

$\Rightarrow \gamma_j \in \{0, 1\}$ identifies the presence of the polygon in the cell complex

Topological Parametric Dictionaries

- A filter acting on edge signals is a polynomial of the Laplacians:

$$\mathbf{H}(\mathbf{L}_1) = h\mathbf{I} + \sum_{l_1=1}^{L_1} h_{l_1, l_1} (\mathbf{L}_{1, \ell})^{l_1} + \sum_{l_2=1}^{L_2} h_{u, l_2} (\mathbf{L}_{1, u})^{l_2}$$

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- Each filter $\mathbf{H}_1 := \mathbf{H}(\mathbf{L}_1)$ is function of its coefficients $h, \{h_{l_1, l_1}\}, \{h_{u, l_2}\}$, and the $\gamma = [\gamma_1, \dots, \gamma_P]^\top$

Topological Parametric Dictionaries

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$\Rightarrow \gamma_j \in \{0, 1\}$ identifies the presence of the polygon in the cell complex

- ▶ Each filter $\mathbf{H}_1 := \mathbf{H}(\mathbf{L}_1)$ is a function of its coefficients $h, \{h_{l,l_1}\}, \{h_{u,l_2}\}$, and the $\gamma = [\gamma_1, \dots, \gamma_P]^\top$
- ▶ The overcomplete topological dictionary is: $\mathbf{D} = [\mathbf{H}_1, \dots, \mathbf{H}_M] \in \mathbb{R}^{N_1 \times MN_1}$
 $\Rightarrow$ function of filter coefficients $\mathbf{h}$ and polygon indicator γ , i.e., $\mathbf{D} = \mathbf{D}(\mathbf{h}, \gamma)$

Topological Dictionary Learning

- ▶ A training set of M flow signals $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_M] \in \mathbb{R}^{N_1 \times M}$

Topological Dictionary Learning

- ▶ A training set of M flow signals $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_M] \in \mathbb{R}^{N_1 \times M}$
- ▶ The problem is cast as the **joint** optimization of the dictionary **coefficients**, the higher-order topological **structure**, and the sparse **signal** representation:

$$(\mathbf{h}^*, \mathbf{t}^*, \mathbf{S}^*) = \arg \min_{\mathbf{h}, \gamma, \mathbf{S}} \|\mathbf{F} - \mathbf{D}(\mathbf{h}, \gamma) \mathbf{S}\|_F^2 + \delta \|\mathbf{h}\|_2^2$$

subject to:

$$\|\mathbf{s}_i\|_0 \leq K_0, \quad i = 1, \dots, M \rightarrow \text{sparsity requirement}$$

$$\mathbf{0} \preceq \mathbf{H}_p \preceq d, \quad p = 1, \dots, S \rightarrow \text{non-negative \& bounded behavior}$$

$$(d - \epsilon) \preceq \sum_{p=1}^P \mathbf{H}_p \preceq (d + \epsilon) \rightarrow \text{whole coverage}$$

$$\gamma_j \in \{0, 1\} \text{ for all } j \rightarrow \text{polygon indicator}$$

$\Rightarrow \mathbf{s}_i$ is the i -th column of $\mathbf{S} \in \mathbb{R}^{PN_1 \times M}$, i.e. the sparse signal representation

- ▶ The problem is non-convex $\Rightarrow$ Approximate iterative solutions

Topological Dictionary Learning

Greedy Topological Dictionary Learning

Repeat until convergence

(S.1): [Dictionary update]:

$$\begin{aligned} \mathbf{h}^{k+1} = \arg \min_{\mathbf{h}} & \left\| \mathbf{F} - \mathbf{D}(\mathbf{h}, \gamma) \mathbf{S}^k \right\|_F^2 + \delta \|\mathbf{h}\|_2^2 \\ & \text{subject to } \mathbf{D}(\mathbf{h}, \gamma) \in \mathcal{D} \end{aligned}$$

Topological Dictionary Learning

Greedy Topological Dictionary Learning

Repeat until convergence

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(S.2): [Sparse Coding]:

$$\begin{aligned} \mathbf{S}^{k+1} &= \arg \min_{\mathbf{S}} \left\| \mathbf{F} - \mathbf{D}(\mathbf{h}^{k+1}, \gamma) \mathbf{S} \right\|_F^2 \\ &\text{subject to: } \|\mathbf{s}_i\|_0 \leq K_0, i = 1, \dots, M \end{aligned}$$

Topological Dictionary Learning

Greedy Topological Dictionary Learning

Repeat until convergence

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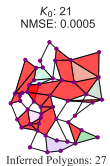
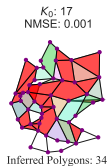
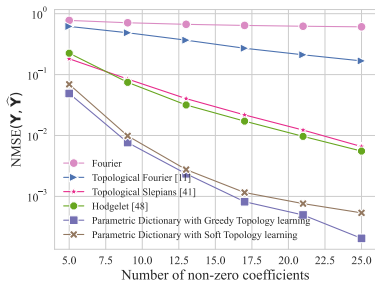
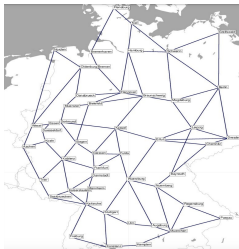
(S.2): [Sparse Coding]:

$$\begin{aligned} \mathbf{S}^{k+1} &= \arg \min_{\mathbf{S}} \left\| \mathbf{F} - \mathbf{D}(\mathbf{h}^{k+1}, \gamma) \mathbf{S} \right\|_F^2 \\ &\text{subject to: } \|\mathbf{s}_i\|_0 \leq K_0, \quad i = 1, \dots, M \end{aligned}$$

(S.3): [Greedy topology update]: Greedy approach starting from $\gamma = \mathbf{1}$

- ▶ Run Algorithm, and then remove the polygon that leads to the largest objective reduction
- ▶ Repeat until there is no more incentive to remove polygons

Results German Network



- **Topology:** German National Research and Education Network (DFN)
- **Edge flows:** aggregated daily data traffic in Mbit/sec and collected on each link
- Topological dictionary learning achieves the best result among competitors

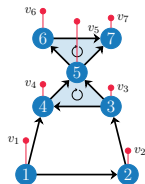
Convolution Across Levels

Recall: Incidence Matrices as Physical Operators

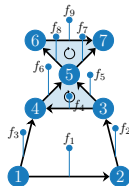
- ▶ Recall the physical meaning of projections using incidence matrices $\mathbf{B}_1$ and $\mathbf{B}_2$:
- ▶ **Node-Edge Incidence $\mathbf{B}_1$**
 - ⇒ **Gradient Flow ($\mathbf{B}_1^\top \mathbf{v}$):** Projects node signals $\mathbf{v}$ onto edges. Calculates the difference in node values across an edge.
 - ⇒ **Divergence ($\mathbf{B}_1 \mathbf{f}$):** Projects edge flows $\mathbf{f}$ onto nodes. Measures the net flow entering or leaving a node.
- ▶ **Edge-Triangle Incidence $\mathbf{B}_2$**
 - ⇒ **Curl Flow ($\mathbf{B}_2 \mathbf{t}$):** Projects triangle signals $\mathbf{t}$ onto edges. Induces a circulating flow along the boundaries of a triangle.
 - ⇒ **Curl ($\mathbf{B}_2^\top \mathbf{f}$):** Projects edge flows $\mathbf{f}$ onto triangles. Measures the net circulation along the triangle's perimeter.

Learn from Joint Signals

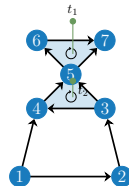
- Encounter joint signals and leverage also **level-to-level** structure



Node signals



Edge signals



Triangle signals

- Simplicial complex signals

$$\mathbf{x} = \begin{bmatrix} \mathbf{v} \\ \mathbf{f} \\ \mathbf{t} \end{bmatrix}$$

⇒ We want to learn from $\mathbf{x}$ and the **whole** simplicial complex

Processing Joint Signals: The Dirac Operator

- To process cross-level signals, we introduce the **Dirac operator**

$$\mathbf{D} = \mathbf{D}_l + \mathbf{D}_u = \begin{bmatrix} \mathbf{0} & \mathbf{B}_1 & \mathbf{0} \\ \mathbf{B}_1^\top & \mathbf{0} & \mathbf{B}_2 \\ \mathbf{0} & \mathbf{B}_2^\top & \mathbf{0} \end{bmatrix}$$

⇒ where lower and upper level interactions are captured by:

$$\mathbf{D}_l = \begin{bmatrix} \mathbf{0} & \mathbf{B}_1 & \mathbf{0} \\ \mathbf{B}_1^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad \mathbf{D}_u = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{B}_2 \\ \mathbf{0} & \mathbf{B}_2^\top & \mathbf{0} \end{bmatrix}$$

⇒ A fundamental property of the Dirac operator is its relation to the Hodge Laplacian:

$$\mathbf{D}^2 = \mathcal{L} = \begin{bmatrix} \mathbf{L}_0 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{L}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{L}_2 \end{bmatrix}$$

Topological Shifts with Dirac

- Simplicial complexes joint signals can be shifted (convolved) by the Dirac operator

$$\begin{bmatrix} \tilde{\mathbf{v}} \\ \tilde{\mathbf{f}} \\ \tilde{\mathbf{t}} \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{B}_1 & \mathbf{0} \\ \mathbf{B}_1^\top & \mathbf{0} & \mathbf{B}_2 \\ \mathbf{0} & \mathbf{B}_2^\top & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{v} \\ \mathbf{f} \\ \mathbf{t} \end{bmatrix} = \begin{bmatrix} \mathbf{B}_1 \mathbf{f} \\ \mathbf{B}_1^\top \mathbf{v} + \mathbf{B}_2 \mathbf{t} \\ \mathbf{B}_2^\top \mathbf{f} \end{bmatrix}$$

- Notice how the different levels interact physically:
 - ⇒ **Nodes** receive divergence from edges ($\mathbf{B}_1 \mathbf{f}$)
 - ⇒ **Edges** receive gradient from nodes and curl from triangles ($\mathbf{B}_1^\top \mathbf{v} + \mathbf{B}_2 \mathbf{t}$)
 - ⇒ **Triangles** receive boundary flows from edges ($\mathbf{B}_2^\top \mathbf{f}$)

Architectures for Joint Filtering (1/2)

- ▶ How do we build neural networks on top of these joint cross-level signals?
 - ⇒ The first main paradigm is **Direct Dirac Filtering**.

- ▶ Apply polynomials of the Dirac operator directly to the joint signal $\mathbf{x}$:

$$\mathbf{H}(\mathbf{D}) = \sum_{j=0}^J h_{j,l} \mathbf{D}_l^j + \sum_{j=0}^J h_{j,u} \mathbf{D}_u^j$$

- ▶ **Key Characteristics:**
 - ⇒ The Dirac operator naturally connects adjacent dimensions.
 - ⇒ This approach mixes signals across all topological levels (nodes, edges, triangles) **simultaneously** at every single layer of the network.

Architectures for Joint Filtering (2/2)

► 2. Decoupled Convolve-and-Project [e.g., Kalman paper framework]

- Instead of a single Dirac polynomial, we define a block-structured joint filter:

$$\mathbf{H}(\mathbf{L}) = \begin{bmatrix} \mathbf{H}(\mathbf{L}_0) & \mathbf{B}_1 \mathbf{H}(\mathbf{L}_1) & \mathbf{0} \\ \mathbf{B}_1^\top \mathbf{H}(\mathbf{L}_0) & \mathbf{H}(\mathbf{L}_1) & \mathbf{B}_2 \mathbf{H}(\mathbf{L}_2) \\ \mathbf{0} & \mathbf{B}_2^\top \mathbf{H}(\mathbf{L}_1) & \mathbf{H}(\mathbf{L}_2) \end{bmatrix}$$

► Diagonal Blocks (Intra-level):

⇒ Independent spatial convolution within nodes, edges, or triangles.

► Off-diagonal Blocks (Cross-level):

⇒ *Convolve then Project*. (e.g., $\mathbf{B}_1^\top \mathbf{H}(\mathbf{L}_0)$ filters nodes, then projects the gradient onto edges).

► Advantage:

⇒ Highly interpretable; explicitly decouples spatial filtering from cross-dimensional flow.

Processing Joint Signals

- Dirac operator and the simplicial complexes signals

$$\mathbf{D} = \mathbf{D}_l + \mathbf{D}_u = \begin{bmatrix} \mathbf{0} & \mathbf{B}_1 & \mathbf{0} \\ \mathbf{B}_1^\top & \mathbf{0} & \mathbf{B}_2 \\ \mathbf{0} & \mathbf{B}_2^\top & \mathbf{0} \end{bmatrix}$$

$\Rightarrow$ where

$$\mathbf{D}_l = \begin{bmatrix} \mathbf{0} & \mathbf{B}_1 & \mathbf{0} \\ \mathbf{B}_1^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad \mathbf{D}_u = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{B}_2 \\ \mathbf{0} & \mathbf{B}_2^\top & \mathbf{0} \end{bmatrix}$$

$\Rightarrow$ The Dirac operator satisfying

$$\mathbf{D}^2 = \mathcal{L} = \begin{bmatrix} \mathbf{L}_0 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{L}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{L}_2 \end{bmatrix}$$

Topological filters with Dirac

- ▶ Simplicial complexes signals can be shifted by the Dirac operator

$$\begin{bmatrix} \tilde{\mathbf{v}} \\ \tilde{\mathbf{f}} \\ \tilde{\mathbf{t}} \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{B}_1 & \mathbf{0} \\ \mathbf{B}_1^\top & \mathbf{0} & \mathbf{B}_2 \\ \mathbf{0} & \mathbf{B}_2^\top & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{v} \\ \mathbf{f} \\ \mathbf{t} \end{bmatrix} = \begin{bmatrix} \mathbf{B}_1 \mathbf{f} \\ \mathbf{B}_1^\top \mathbf{v} + \mathbf{B}_2 \mathbf{t} \\ \mathbf{B}_2^\top \mathbf{t} \end{bmatrix}.$$

- ▶ Dirac filters

$$\mathbf{H} = \underbrace{\sum_{j=1}^J h_{j,l} \mathbf{D}_l^j}_{\mathbf{H}_l} + \underbrace{\sum_{j=1}^J h_{j,u} \mathbf{D}_u^j}_{\mathbf{H}_u} + \underbrace{h_h \tilde{\mathbf{Q}}}_{\mathbf{H}_h}$$

$\Rightarrow \tilde{\mathbf{Q}}$ approximates the orthogonal projector onto the harmonic subspace

Further Reading

- ▶ Yang, Maosheng, Elvin Isufi, Michael T. Schaub, and Geert Leus. "Simplicial convolutional filters." *IEEE Transactions on Signal Processing* 70 (2022): 4633-4648.
- ▶ Isufi, Elvin, and Maosheng Yang. "Convolutional filtering in simplicial complexes." *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022.
- ▶ Grimaldi, Enrico, Claudio Battiloro, and Paolo Di Lorenzo. "Topological dictionary learning." *IEEE Transactions on Signal Processing* (2025).
- ▶ Calmon, Lucille, Michael T. Schaub, and Ginestra Bianconi. "Dirac signal processing of higher-order topological signals." *New Journal of Physics* 25.9 (2023): 093013.
- ▶ Money, R., Sabbaqi, M., Krishnan, J., Beferull-Lozano, B., Isufi, E. "Kalman filtering for simplicial processes." *2024 58th Asilomar Conference on Signals, Systems, and Computers*. IEEE, 2024.
- ▶ Liu, C., Tenorio, V. M., Marques, A. G., Isufi, E. (2026). *Matched Topological Subspace Detector*. *IEEE Transactions on Signal Processing*.

Video material

- ▶ https://www.youtube.com/watch?v=F_HSgevvvzg
- ▶ <https://www.youtube.com/watch?v=Ylwz6BWeIvA>
- ▶ <https://www.youtube.com/watch?v=PPy8SJpmkCU>